

RISK MANAGEMENT

1. Risk Management Approach

The project will follow a **PMBOK 8 uncertainty-based risk management approach**, where both **threats and opportunities** are identified and managed to maximize project value.

- Risks will be identified using **TECOP (Technical, External, Commercial, Organisational, Project Management)** categories
- Risk analysis will include **qualitative (Probability × Impact)** and basic **quantitative (EMV)** techniques
- Each risk will have a **defined owner and response strategy**
- Risks will be continuously monitored throughout the project lifecycle
- A **risk register** will be maintained and updated regularly

2. Risk Identification:

R1 – Technical

Because the AI model may not fully understand uploaded content, there is a risk that **irrelevant or incorrect questions are generated**, which could result in **poor quiz quality and reduced user trust**.

R2 – Technical

Because the system may face high concurrent usage, there is a risk that **performance issues or system crashes occur**, which could result in **delays and failed quiz attempts**.

R3 – External

Because the project depends on third-party AI APIs, there is a risk that **API service becomes unavailable or changes**, which could result in **AI module failure**.

R4 – Organisational

Because team members have limited experience with AI integration, there is a risk that **development delays occur**, which could result in **missing deadlines**.

R5 – Project Management

Because requirements may evolve during development, there is a risk that **scope creep occurs**, which could result in **increased workload and timeline delays**.

OPP1 – Opportunity

Because AI integration is innovative, there is an opportunity that the system **gains high academic recognition**, which could result in **better grades and future project scalability**.

3. Risk Analysis

3.1 Qualitative Analysis (P × I)

ID	Risk	P	I	Score	Priority
R1	AI inaccurate output	0.6	0.7	42	Critical
R2	System performance issues	0.5	0.8	40	Critical
R3	API failure	0.4	0.7	28	High
R4	Team skill gap	0.5	0.6	30	High
R5	Scope creep	0.6	0.5	30	High
OPP1	Academic recognition	0.4	0.8	32	Opportunity

3.2 Quantitative Analysis (EMV)

Risk	P	Impact (PKR)	EMV
R1	0.6	-30,000	-18,000
R2	0.5	-50,000	-25,000
R3	0.4	-40,000	-16,000
OPP1	0.4	+60,000	+24,000

Total EMV ≈ -35,000 PKR

Recommended **contingency reserve ≈ 35K PKR**

4. Risk Response Planning:

4.1 Threat Strategies (4Ts)

R1 – AI Incorrect Questions → MITIGATE

- Add manual editing feature
- Test AI output before publishing
- **Owner:** AI Developer
- **Trigger:** >20% incorrect questions

R2 – System Performance Issues → MITIGATE

- Optimize backend
- Use scalable services (Firebase)
- **Owner:** Backend Developer
- **Trigger:** Slow response > 3 sec

R3 – API Failure → TRANSFER + MITIGATE

- Use backup API / fallback logic
- Cache generated questions
- **Owner:** Developer
- **Trigger:** API downtime

R4 – Skill Gap → MITIGATE

- Use tutorials, documentation
- Divide tasks based on expertise
- **Owner:** Project Manager

R5 – Scope Creep → AVOID

- Freeze requirements after approval
- Use MoSCoW prioritization
- **Owner:** Project Manager

4.2 Opportunity Strategy

OPP1 – Academic Recognition → EXPLOIT

- Focus on AI feature quality
- Improve UI/UX for presentation
- **Owner:** Project Manager

5. Contingency & Fallback Plans

Risk: AI generates poor questions

- **Contingency Plan:**
 - Allow teachers to edit questions
 - Use predefined question bank
- **Fallback Plan:**
 - Disable AI feature and use manual quiz creation only

6. Risk Monitoring & Control

- Weekly risk review meetings
- Update risk register regularly
- Track risk status (Open / Closed / Mitigated)
- Identify new risks during development
- Reassess probability and impact

7. Risk Register:

ID	Risk	Cat.	P	I	Score	Owner	Strategy	Status
R1	AI inaccurate output	Tech	0.6	0.7	42	AI Dev	Mitigate	Open
R2	Performance issues	Tech	0.5	0.8	40	Backend Dev	Mitigate	Open
R3	API failure	External	0.4	0.7	28	Dev	Transfer	Open
R4	Skill gap	Org	0.5	0.6	30	PM	Mitigate	Open

ID	Risk	Cat.	P	I	Score	Owner	Strategy	Status
R5	Scope creep	PM	0.6	0.5	30	PM	Avoid	Open
OPP1	Academic success	PM	0.4	0.8	32	Team	Exploit	Open